

Technical Specifications

Test Bed and Instrument Frame

Item	Specification
Operating Environment	Indoor (laboratory) Altitude up to 2000 m Overvoltage category 2 (as specified in EN61010-1). Pollution degree 2 (as specified in EN61010-1).
Test Bed (with Dynamometer and Coupling Guard)	
Length	960 mm
Height	1000 mm
Depth	480 mm
Net Weight	110 kg
Water Supply for Dynamometer	Constant head Minimum 1 bar pressure at 5 Litres per minute
Maximum Dynamometer Rating	7.5 kW at 7000 rev.min ⁻¹
Instrument Frame	
Length	1260 mm
Height	800 mm
Depth	520 mm
Net Weight	24.5 kg
Supply Voltage	90 VAC to 250 VAC Single Phase 50 Hz to 60 Hz
Supply Rating	10 A Maximum Load
Fuse	20 mm 10 A Type T

Instrument Units

Item	Specification
Volumetric Fuel Gauge - AVF1	Gauge Weight: 2.5 kg Fuel Capacities: 8 mL, 16 mL and 24 mL
Engine Inlet Air and Exhaust Display - DPT1	1 x Differential Pressure Transducer 1 x Built-in Ambient Pressure Transducer 2 x (Extra Low Voltage <25 VDC) Sockets for K-type thermocouples (Engine air Inlet and Test Engine Exhaust) 1 x RJ45 type socket for VDAS connection (Extra Low Voltage <25 VDC) Display Module Supply: 90 VAC to 250 VAC 50 Hz to 60 Hz 100 mA. F6.3 A Fuse Module Weight: 4 kg Module Size: 190 mm x 125 mm x 450 mm

Item	Specification
Torque and Speed Display - DTS2	2 x (Extra Low Voltage <25 VDC) DIN type sockets for Dynamometer Torque and Speed 1 x RJ45 type socket for VDAS connection (Extra Low Voltage <25 VDC) Display Module Supply: 90 VAC to 250 VAC 50 Hz to 60 Hz 100 mA. F6.3 A Fuse Module Weight: 4.5 kg Module Size: 190 mm x 125 mm x 450 mm
Test Bed Instruments	K Type Thermocouple at Air Inlet Orifice Diameter: Nominal 18.5 mm - actual value marked on equipment Dynamometer Torque (Load Cell/Force sensor) Dynamometer Speed (Optical Sensor)

TDX00a Optional Gas Calorimeter

Item	Specification
Dimensions	
Calorimeter	930 mm (length), 300 mm (height), 355 mm (depth), 16 kg (weight)
Water Valve	250 mm (length), 50 mm (height), 110 mm (depth), 1 kg (weight)
MCSS	
Pressure Relief	4 bar
Water Supply	10 l.m ⁻¹ @ 1 bar minimum, 3 bar maximum
Control Box	
Dimensions	190 mm (width), 450 mm (height), 130 mm (depth), 4 kg (weight)
Fuse	F6.3A
Connections	4 x thermocouple socket, 1 x flowmeter socket
Electrical Supply	Comes from TD Instrument Console

Properties of Fuels

Fuel	Lower Calorific Value (MJ.kg ⁻¹)	Stoichiometric air/fuel ratio	Density (kg.L ⁻¹)	Density (kg.m ⁻³)
Petrol/Gasoline	43.8	14.6	0.74	740
Diesel	39		0.84	840
Light Fuel Oil	40.6		0.925	925
Medium Fuel Oil	39.9	14.4	0.95	950
Heavy Fuel Oil	39.7		0.965	965

Table 2 Properties of Fuels

Noise Levels

Refer to the TecQuipment User Guide supplied with each Test Engine.